

Steps

Reason

1. $\forall x(P(x) \vee Q(x))$
2. $\forall x(\neg Q(x) \vee S(x))$
3. $\forall x(R(x) \rightarrow \neg S(x))$
4. $\exists x \neg P(x)$
5. $\neg P(c)$ for some c in the domain

- Premise
Premise
Premise
Premise
Existential instantiation on 4

6. $P(c) \vee Q(c)$
7. $\neg Q(c) \vee S(c)$
8. $Q(c) \vee P(c)$
9. $P(c) \vee S(c)$
10. $S(c)$
11. $R(c) \rightarrow \neg S(c)$
12. $\neg(\neg S(c))$
13. $\neg R(c)$
14. $\exists x \neg R(x)$

- Universal instantiation on 1
Universal instantiation on 2
Commutative law on 6
Resolution on 8, 7
Disjunctive syllogism on 9, 5
Universal instantiation on 3
Double negation law on 10
Modus tollens on 12, 11
Existential generalization on 13

30. Use resolution to show the hypotheses "Allen is a bad boy or Hillary is a good girl." and "Allen is a good boy or David is Happy" imply the conclusion "Hillary is a good girl or David is happy."

Soln: P: Allen is a good boy Q: Hillary is a good girl
R: David is happy

Premise: 1. $\neg P \vee Q$ 2. $P \vee R$ Conclusion: $Q \vee R$

Clause C1: $\neg P \vee Q$
Clause C2: $P \vee R$
Clause C3: $\neg(Q \vee R) \equiv \neg Q \wedge \neg R$ (By De Morgan's law)
By Resolution on C2, C1, we get C4: $R \vee Q$
By Commutative law, C5: $Q \vee R$